

DIAGONAL-FEEDBACK COMMUTATORS AND SUPPORT-COLOURING FIBRES

ANONYMOUS

ABSTRACT. Let $K = \mathbb{F}_q$ and iterate the state-feedback commutator $\Phi(A) = [\text{Diag}(a_{11}, \dots, a_{nn}), A]$ on $M_n(K)$, with the standard ordered basis fixed. The map satisfies $\Phi^2 = 0$, but its one-step inverse problem is nonuniform. We prove that a matrix B is reachable precisely when its diagonal is zero and the undirected support graph of its off-diagonal entries is q -colourable. More strongly, every target fibre is an occupation-weighted proper-colouring sum. This gives closed graph sums for the image, a weak-composition formula for the kernel, and the complete functional graph: one fixed root, explicit depth-one leaves and branch vertices, and support-indexed depth-two leaves. The zero matrix uniquely maximizes the one-step fibre, and all later fibres and image sets are immediate. Classical matrix and group commutator theory and Potts-chromatic partition functions are treated as prior owner regions; the residual claim concerns only their exact conjunction for this literal self-map. External circulation remains on hold.

1. DIAGONAL FEEDBACK AND SCOPE

Fix a prime power q , put $K = \mathbb{F}_q$, and let

$$(1) \quad \Delta(A) = \text{Diag}(a_{11}, \dots, a_{nn}), \quad \Phi : M_n(K) \longrightarrow M_n(K), \quad \Phi(A) = [\Delta(A), A].$$

The ordered basis is part of the definition: neither Δ nor Φ is similarity invariant. Entrywise,

$$(2) \quad \Phi(A)_{ij} = (a_{ii} - a_{jj})a_{ij}.$$

Thus the diagonal is erased after one step. The temporal clock is therefore short, yet (2) leaves a nonconstant inverse multiplicity which records how the input diagonal partitions the coordinate set.

This note makes a deliberately narrow claim. Additive commutators with prescribed diagonal behaviour belong to established matrix-commutator theory [4, 7]; images and fibres of classical commutator word maps have their own literature [2, 5]. Fixed-element triangular Engel equations are treated by Bier [3]. In particular, the fixed-regular unitriangular commutator system catalogued as P119 in the accompanying programme—including its centralizer-coset fibres and filtration tree—is assigned no contribution credit here. Likewise, proper colourings and Potts/Tutte partition functions are prior combinatorial mechanisms [6]. What remains is the exact image-fibre-clock package forced by the state-dependent diagonal in (1); no novelty or priority conclusion is drawn from the bounded literature audit.

For a zero-diagonal $B \in M_n(K)$, let G_B be the simple graph on $[n] = \{1, \dots, n\}$ in which

$$(3) \quad \{i, j\} \in E(G_B) \iff b_{ij} \neq 0 \text{ or } b_{ji} \neq 0.$$

For a colouring $c : [n] \rightarrow K$, write

$$n_\alpha(c) = |c^{-1}(\alpha)|, \quad m(c) = \sum_{\alpha \in K} n_\alpha(c)(n_\alpha(c) - 1).$$

Here $m(c)$ counts ordered pairs of distinct vertices of equal colour. For a simple graph G define the occupation-marked proper-colouring sum

$$(4) \quad \mathcal{P}_{G,q}(X; \mathbf{z}) = \sum_{c \in \text{Col}_q(G)} X^{m(c)} \prod_{\alpha \in K} z_\alpha^{n_\alpha(c)},$$

where $\text{Col}_q(G)$ is the set of proper K -colourings of G .

2. SUPPORT COLOURINGS AND EVERY-TARGET FIBRES

Theorem 1 (complete one-step fibre). *For every $B \in M_n(K)$,*

$$(5) \quad |\Phi^{-1}(B)| = \begin{cases} \mathcal{P}_{G_B,q}(q; \mathbf{1}), & \text{diag}(B) = 0, \\ 0, & \text{diag}(B) \neq 0. \end{cases}$$

More precisely, for every occupation vector $\mathbf{r} = (r_\alpha)_{\alpha \in K}$, the number of sources in the fibre with $|\{i : a_{ii} = \alpha\}| = r_\alpha$ is the coefficient of $\mathbf{z}^{\mathbf{r}}$ in $\mathcal{P}_{G_B,q}(q; \mathbf{z})$. Consequently,

$$(6) \quad B \in \text{im } \Phi \iff \text{diag}(B) = 0 \text{ and } \chi(G_B) \leq q,$$

and the fibre cardinality depends only on the undirected support G_B .

Proof. Every output has zero diagonal by (2), proving the second line of (5). Suppose now that B has zero diagonal and freeze the prospective input diagonal as $c_i = a_{ii}$. For each ordered pair $i \neq j$ we must solve

$$(7) \quad (c_i - c_j)a_{ij} = b_{ij}.$$

If $c_i \neq c_j$, there is exactly one solution. If $c_i = c_j$, there are q solutions when $b_{ij} = 0$ and none when $b_{ij} \neq 0$. Hence the diagonal c is admissible exactly when it is a proper colouring of G_B . For such a colouring, precisely $m(c)$ ordered off-diagonal entries are free, so it contributes $q^{m(c)}$ sources. Summing over c proves (5); retaining the monomial in (4) proves the marked assertion. Positivity of every summand then gives (6) and support-only dependence. $\square$

Write $\kappa_{n,q} = |\Phi^{-1}(0)|$ and $I_{n,q} = |\text{im } \Phi|$.

Corollary 2 (kernel, image, and maximum). *The zero matrix uniquely maximizes $|\Phi^{-1}(B)|$. Moreover,*

$$(8) \quad \kappa_{n,q} = \sum_{\substack{(r_\alpha)_{\alpha \in K} \geq 0 \\ \sum_{\alpha} r_\alpha = n}} \binom{n}{(r_\alpha)_{\alpha \in K}} q^{\sum_{\alpha} r_\alpha (r_\alpha - 1)},$$

$$(9) \quad I_{n,q} = \sum_{\substack{G \subseteq K_n \\ \chi(G) \leq q}} (q^2 - 1)^{|E(G)|}.$$

The graphs in (9) are simple and labelled on $[n]$.

Proof. For $B = 0$, the support graph is empty, so every colouring contributes; group these colourings by their labelled occupation vector to obtain (8). If $B \neq 0$ has zero diagonal, then G_B contains an edge. Its proper colourings form a strict subset of all colourings, while all weights $q^{m(c)}$ are positive. If B has nonzero diagonal, its fibre is empty. Thus zero is the unique maximum.

For (9), first choose the support G . It occurs precisely when G is q -colourable. On every edge $\{i, j\}$, the ordered pair (b_{ij}, b_{ji}) may be any nonzero vector in K^2 , giving $q^2 - 1$ choices, independently across edges. Nonedges force both entries to zero. $\square$

The quantity in (4) is usefully viewed as a Potts-type proper-colouring partition sum with a global occupation interaction. That terminology does not identify it with the ordinary chromatic polynomial or claim ownership of the Potts formalism.

3. THE COMPLETE FUNCTIONAL GRAPH

Theorem 3 (clock, tree, and all-time fibres). *The map satisfies $\Phi^2 = 0$. Its functional graph is one rooted component with the zero matrix as its only periodic point and with a loop at the root. Its sharp height is 1 for $n = 1$ and for $n \geq 2$. The depth layers have cardinalities*

$$(10) \quad D_0 = 1, \quad D_1 = \kappa_{n,q} - 1, \quad D_2 = q^{n^2} - \kappa_{n,q},$$

where $D_2 = 0$ when $n = 1$.

Among the depth-one vertices, $I_{n,q} - 1$ are branch vertices and $\kappa_{n,q} - I_{n,q}$ are leaves. For every nonempty q -colourable labelled graph G , there are $(q^2 - 1)^{|E(G)|}$ branch vertices with support G , and each has $\mathcal{P}_{G,q}(q; \mathbf{1})$ depth-two leaf preimages. Equivalently,

$$(11) \quad \sum_{\substack{\emptyset \neq G \subseteq K_n \\ \chi(G) \leq q}} (q^2 - 1)^{|E(G)|} \mathcal{P}_{G,q}(q; \mathbf{1}) = q^{n^2} - \kappa_{n,q}.$$

For $t \geq 0$ and any target B ,

$$(12) \quad |(\Phi^t)^{-1}(B)| = \begin{cases} 1, & t = 0, \\ |\Phi^{-1}(B)|, & t = 1, \\ q^{n^2}, & t \geq 2 \text{ and } B = 0, \\ 0, & t \geq 2 \text{ and } B \neq 0. \end{cases}$$

In particular, $\text{im } \Phi^0 = M_n(K)$, $|\text{im } \Phi| = I_{n,q}$, and $\text{im } \Phi^t = \{0\}$ for $t \geq 2$.

Proof. Every $C = \Phi(A)$ has zero diagonal. Therefore $\Delta(C) = 0$ and $\Phi(C) = [0, C] = 0$, proving $\Phi^2 = 0$. It follows that every orbit enters zero within two steps and that $\text{im } \Phi \subseteq \ker \Phi$. A fixed point A obeys $A = \Phi(A) = \Phi^2(A) = 0$; the same collapse excludes every other periodic orbit.

The zero fibre is the kernel of the set map, of size $\kappa_{n,q}$, so it consists of the root and $\kappa_{n,q} - 1$ nonzero depth-one vertices. All remaining $q^{n^2} - \kappa_{n,q}$ states have nonzero image and hence depth two, proving (10). For $n = 1$, every output is zero. If $n \geq 2$, choose two diagonal entries of different colours and one off-diagonal entry nonzero; its image is nonzero, so height two occurs.

A nonzero depth-one vertex has predecessors exactly when it lies in the image. Thus the $I_{n,q} - 1$ nonzero image points are precisely the branch vertices, leaving $\kappa_{n,q} - I_{n,q}$ leaves. The support census from Corollary 2, followed by Theorem 1, gives the branching statement. Every predecessor of a nonzero image point has depth two and cannot itself lie in the image, so these predecessors are leaves. Summing them proves (11). Finally Φ^2 is the constant-zero map, which gives (12) and the image tower. $\square$

Because every iterate has exactly one fixed point, the Artin–Mazur bookkeeping [1] gives

$$(13) \quad \zeta_{\Phi}(z) = \exp\left(\sum_{r \geq 1} \frac{|\text{Fix}(\Phi^r)|}{r} z^r\right) = \frac{1}{1-z}.$$

4. EXACT CONTROLS AND LIMITATIONS

An independent standard-library verifier enumerates every literal arrow and every codomain target in eleven boxes. It evaluates the marked fibre sum, tests support-only dependence and image membership, independently enumerates support graphs for (9), evaluates (8), and checks the depth layers. The field implementation includes $\mathbb{F}_4 = \mathbb{F}_2[x]/(x^2 + x + 1)$, so the controls are not confined to prime fields. Representative boxes are shown below; $S = q^{n^2}$.

n	q	S	$I_{n,q}$	$\kappa_{n,q}$	height
1	4	4	1	4	1
2	4	256	16	76	2
3	2	512	37	152	2
3	4	262144	4096	16984	2
4	2	65536	829	8800	2

The settled transcript records 2,111,465 assertions and a deterministic literal-edge digest. These finite calculations are falsification controls, not a substitute for the uniform proofs. The dynamics has a deliberately shallow clock, while its inverse axis lies in an owner-dense colouring region. External circulation, novelty, priority, and submission therefore remain `HOLD_EXTERNAL` pending a later hostile review.

REFERENCES

- [1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
- [2] Robert W. Baddeley. Images of commutator maps. *Communications in Algebra*, 22(8):3023–3035, 1994. doi: 10.1080/00927879408825010.
- [3] Agnieszka Bier. On solvability of Engel equations in the group of triangular matrices over a field. *Linear Algebra and its Applications*, 438(5):2320–2330, 2013. doi: 10.1016/j.laa.2012.10.009.
- [4] Zhibek Kadyrsizova and Madi Yerlanov. Algebraic sets defined by the commutator matrix. *Journal of Algebra*, 589:29–50, 2022. doi: 10.1016/j.jalgebra.2021.09.012.
- [5] Michael Larsen and Zhipeng Lu. Flatness of the commutator map over SL_n . *International Mathematics Research Notices*, 2021(8):5605–5622, 2021. doi: 10.1093/imrn/rnz285.
- [6] Alan D. Sokal. The multivariate Tutte polynomial (alias Potts model) for graphs and matroids. In Bridget S. Webb, editor, *Surveys in Combinatorics 2005*, volume 327 of *London Mathematical Society Lecture Note Series*, pages 173–226. Cambridge University Press, 2005. doi: 10.1017/CBO9780511734885.009.
- [7] Hsu-Wen Vincent Young. On matrix pairs with diagonal commutators. *Journal of Algebra*, 570:437–451, 2021. doi: 10.1016/j.jalgebra.2020.11.023.